

READING ATTENTION AT A DEPTH THAT DOES NOT MATTER: TRANSPORT-INFORMED TOKEN ALLOCATION FOR VISION–LANGUAGE MODELS

Anonymous authors

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ABSTRACT

Vision–language models on high-resolution imagery must decide where to spend a fixed visual-token budget; evidence not given resolution is destroyed before the language model runs. The prevailing remedy is attention-guided cropping: read the model’s own attention over image positions, crop where it peaks, re-encode. Every such method must choose a depth at which to read that map, and published methods hand-pick a single layer or average a fixed block. We show this choice dominates every other design decision, and that the depth these methods read is not the depth at which the image reaches the answer. Masking text-to-image attention layer by layer localises that transport to the first two thirds of the stack: blocking every layer from layer 16 of 28 onward leaves the answer almost unchanged (KL 0.014/0.029) and from layer 20 onward flips essentially no answers. The boundary falls at the same fraction of depth, 0.57, on a 36-layer checkpoint as on the two 28-layer ones, entirely before the band these methods read. Two policies follow from that measurement rather than from tuning. **Depth-Weighted Attention** (DWA) replaces a hand-picked depth with one closed-form ridge solve over the whole depth profile, fitted from about fifty boxed examples. **Adaptive Visual Resolution** (AVR) discards 90% of visual tokens at the boundary, where they are causally inert, and spends the freed compute on encoding resolution. At matched compute DWA beats every published placement rule in all nine pooled comparisons. Across four architectures — native dynamic resolution, a fixed-token encoder, dynamic tiling and anyres — depth re-weighting beats the published single-block read-out on four of five checkpoints, by +10.5 to +16.2 points. On Gemma-3 the incumbent is indistinguishable from random ($1.1\times$ chance coverage, +0.5 points) while re-weighting the *same* attention is worth +16.2. AVR converts a checkpoint’s resolution headroom at rate 1.01 ($r=0.966$). Against models at their *native* dynamic resolution — $5.5\times$ and $7.2\times$ the budget — DWA is statistically indistinguishable on both using 14–18% of the visual-token compute, whereas every published rule falls 9.9–14.2 points short.

1 INTRODUCTION

A vision–language model asked “what material is the glove?” about a 2250×1500 photograph must first encode the image into visual tokens. At any finite budget the glove may occupy a fraction of a token, and the evidence is gone before the language model runs. The model still *localises* it — its attention concentrates on the right region — but it cannot *perceive* it.

The standard remedy is attention-guided cropping: run the model once, read its attention over image positions, crop where the attention peaks, and re-encode that crop at higher density (Zhang et al., 2025; 2023; Shen et al., 2024). Every method in this family faces one decision that is rarely examined: **at which layer should the attention map be read?** Published methods pick a single mid-network layer or average a fixed block of late layers.

We show three things.

(i) **The depth choice dominates.** We crop at every one of the 28 layers’ arg-max in turn and answer, on both checkpoints (Figure 2). End-task accuracy ranges from 34.6% to 70.2% (Qwen3) and 36.1% to 62.8% (Qwen2) — spreads of 35.6 and 26.7 points driven by nothing but the depth at which the map is read. Every layer inside the transport window scores below not cropping at all on both, and the published fixed-layer recipe lands 32 and 25 points below the best layer. The same pipeline with a better depth read-out scores 77.2% and 71.3% on the two checkpoints.

(ii) **Everyone reads at the wrong depth, and we can say why.** Masking text tokens’ attention to the image, layer by layer, shows that image-to-text transport is *complete* by layer 16 of 28: blocking every layer from 16 onward leaves the output nearly unchanged (KL 0.014/0.029; 12%/5% flips) and from layer 20 onward flips essentially none. Yet every method in this family — including ours — reads its map at layers 17–22. The attention these methods consume is a residue of where the model looked, not the channel carrying the evidence.

(iii) **Two allocation policies follow from that measurement**, with complementary strengths, and at native resolution DWA matches a model spending $5.5\text{--}7.2\times$ the compute. **Depth-Weighted Attention** (DWA, §6.1) replaces a hand-picked depth with a single closed-form ridge solve over all of them. We test what makes it work rather than asserting it: the conventional read-out fails because of a component of the map that does not depend on the image, and the fit’s job is to learn which handful of depths to read instead (§J). **Adaptive Visual Resolution** (AVR, §6.2) drops 90% of visual tokens at the boundary, where they no longer carry the image, and spends the saved compute on encoding resolution instead.

2 RELATED WORK

Attention-guided cropping. Zhang et al. (2025) establish the premise this family rests on — a model’s attention concentrates on the right region even when its answer is wrong — and turn it into a training-free crop, evaluated over seven VQA datasets; Liu et al. (2026) probe the same disconnect. ViCrop (Zhang et al., 2023; 2025) crops at the peak of a fixed mid-layer attention map; ZoomEye (Shen et al., 2024) searches a tree of crops with repeated model queries. VisLens (He et al., 2026) reaches the same place without attention at all, decoding visual tokens through a tuned logit lens, matching them against the query, and feeding the crop back alongside the original image in one pass. Every attention-based member fixes the read-out depth by hand, and none of them budget-match against simply spending the same tokens on resolution — Table 2 shows that at equal compute the family is worth -1.6 to $+3.1$ points.

Layer-adaptive read-outs. LASER (Zhu et al., 2026) selects the attention layer per sample by contrasting the map with and without the query, and reports that no single “magic layer” suffices — a conclusion we reach independently and extend causally. ILVAD (Xie et al., 2026) accumulates newly-activated tokens across adjacent layers; TRACE (Ye et al., 2026) identifies a three-stage depth pattern and reshapes it with trained modules; ViRGo (Tran et al., 2026) reports our cross-instance finding (“patch-based zooming ... can split large objects and destroy global spatial context”) and resolves it with a three-way router keyed on object scale. We treat these as concurrent work reaching compatible conclusions. Our contribution relative to them is (i) the causal localisation of the transport window rather than a correlational choice of layer, (ii) a *supervised* read-out fitted over all depths rather than a rule chosen by hand, and (iii) budget-matched, stratified evaluation including against native resolution.

Depth selection in language models. DoLa (Chuang et al., 2024) contrasts a mature and a premature layer; SLED (Zhang et al., 2024) weights all early layers by gradient alignment; confident-layer decoding stops at an entropy valley near the output. We port each to this setting and all fail (§8) for one measurable reason: in these vision–language models final-layer entropy rises on 54–88% of items — far above the 16% reported for language models — yet per-layer accuracy still climbs monotonically to the last layer, so on the *decoding* axis there is no better layer to select. The *read-out* axis is different and we are explicit about it: a well-chosen fixed layer does work (L17 scores 70.2% against a 63.7% no-crop bar), but it cannot be identified without labels, the published choice is 31.9 points worse than it, and per-sample selection (LASER) reaches only 64.7%. Re-weighting matches the oracle fixed layer without needing to know it.

Visual-token pruning. FastV (Chen et al., 2024) prunes at layer 2; PyramidDrop (Xing et al., 2024) drops tokens in stages with a fixed ratio; HiRED (Arif et al., 2025) allocates a budget before the language model. These are evaluated as retention-at-reduced-cost. We derive the pruning depth from the transport measurement rather than tuning it, and show that at that depth the *choice* of surviving tokens is irrelevant — random keep matches attention keep — while one layer earlier it is worth +8.4 and +8.9 points.

3 SETUP

Benchmarks. V*Bench (Wu & Xie, 2024) ($n = 191$; 115 single-instance, 76 cross-instance) and HR-Bench 4K/8K (Wang et al., 2024) ($n = 800$ each; 400 per stratum). Both are public; we build no datasets. Questions are four-way multiple choice, scored by the arg-max over option-letter log-probabilities, so results are deterministic.

The equal-compute bar. A two-pass method that localises at 300 tokens and answers on a crop at 300 tokens spends 600 visual tokens. We therefore compare against `uniform@600`: the same budget spent on one higher-resolution pass. Published work in this area does not budget-match; §7 shows what that hides. Where pruning changes cost mid-network we measure **token-layers** (tokens $\times$ layers traversed); the bar is $600 \times 28 = 16,800$.

Stratification. Every result is reported separately for *single-instance* questions (one object’s property) and *cross-instance* questions (a relation between two). §7.3 shows these behave oppositely and that pooling them conceals it.

Statistics. Paired item bootstrap, 8000 resamples. Learned components use GroupKFold(5) by item $\times$ 3 seeds. A claim must hold on at least two checkpoints or it is reported as rejected.

4 THE DEPTH AT WHICH THE MAP IS READ DOMINATES

Before proposing anything we establish the size of the effect that motivates it. We hold the pipeline fixed — same model, same question, same ~ 600 visual-token budget, same fixed $W=0.25$ window — and vary *only* the layer whose attention arg-max supplies the crop, sweeping all 28 layers on both checkpoints and answering at each.

Three facts follow, and each is measured rather than argued. First, the spread attributable to depth alone is 35.6 points on Qwen3-VL-2B and 26.7 on Qwen2-VL-7B — larger than the gap between any two methods compared in this literature. Second, *every* layer inside the transport window scores below not cropping at all, on both checkpoints: read the map too early and attention-guided cropping is actively harmful. Third, the fixed layer used by published practice is 24–32 points below the best layer, and in the pooled comparison of Table 2 it scores 38.2% against a 63.9% no-crop bar — cropping at a hand-picked depth is worse than not cropping.

This is the decision the family does not examine, and it dominates every other design choice in it. The rest of the paper asks why the good depths are where they are (§5), and what follows from knowing (§6).

5 WHERE DOES THE IMAGE ACTUALLY REACH THE ANSWER?

We install a hook that masks a chosen set of query positions from attending to the image-token columns, at chosen layers only, and measure the KL divergence at the output.

Table 1: Causal masking of text-to-image attention on both checkpoints, $n = 40$ each. Qwen3-VL-2B values first.

mask scope (all layers)	KL at output	answers flipped
the answer token ’s row only	0.025 / 0.034	2.5% / 0%
all text rows after the image	1.266 / 0.532	42% / 45%

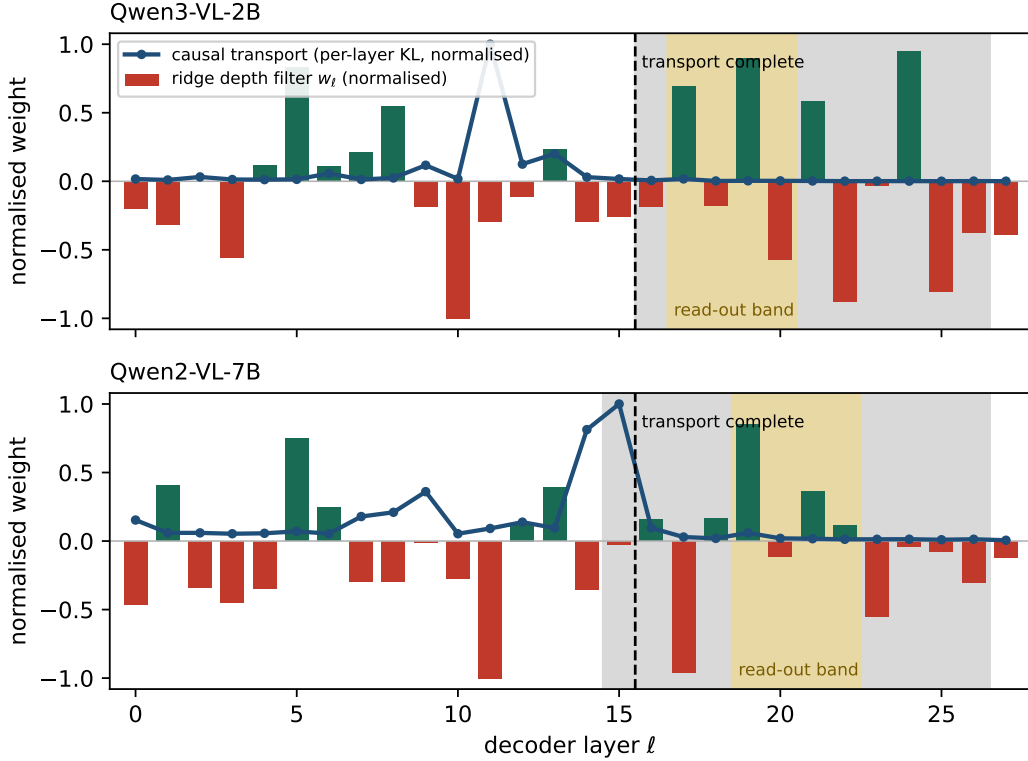


Figure 1: **The fitted depth filter sits after the transport window.** Bars: DWA’s weights w_ℓ (green positive, red negative), normalised. Line: the causal transport profile — the output KL caused by masking text-to-image attention at layer ℓ alone. Grey band: the layers the conventional block-mean read-out averages at $+1$. Yellow band: the question-conditioned read-out band. Dashed line: the measured transport boundary, beyond which masking *all* remaining layers leaves the answer nearly unchanged (12%/5% flips at 16, essentially none from 20). The filter assigns negative weight to seven of the eleven block-mean layers here, and to seven of twelve on the other checkpoint; the band’s net weight is $+0.0006$ and -0.0005 , i.e. approximately zero, against the $+1$ per layer a block mean applies. We do *not* claim the positive mass sits after the boundary: that holds on this checkpoint and reverses on the other, so by the two-model rule of §A it is not a result.

Blocking the answer-emission token from seeing any image token, at every layer, changes almost nothing on either checkpoint (Table 1). Visual evidence reaches the answer through the *text* tokens, which consolidated it upstream. Masking one layer at a time for all text rows localises that consolidation: it peaks at L11 (Qwen3) and L14–15 (Qwen2), prefix masks $L_{0..\ell}$ saturate by $\ell = 12$ –16, and suffix masks $L_{\ell..end}$ fall to ≈ 0 from $\ell = 16$ (0.014 and 0.029; essentially zero answer flips from $\ell = 20$). The same fraction transfers to a 36-layer checkpoint under the same mask: its suffix is already nearly harmless at L20 ($0.56\times$ depth; KL 0.052, 5% flips) and fully harmless from L24, and pruning at L21 — the same fraction — is free (-7.9 points when the cut is instead placed at $0.14\times$ depth).

A second, independent causal test. Masking shows the text-to-image attention *channel* is inert after the boundary. It does not show the image tokens’ *values* are. We therefore also replace the image tokens’ hidden states at layer ℓ with those computed from a *different image* and measure the output (Appendix I). The effect falls by $17\times$ and $96\times$ between L12 and L16 and reaches KL $\approx 5 \times 10^{-4}$ with 0–2% answer flips by L20, while two pre-registered controls stay large: patching image tokens at L4, and patching *text* tokens at L20. At L4 the image matters $151\times$ and $131\times$ more than the text; at L20 the text matters $2492\times$ and $965\times$ more than the image. Two different interventions, two checkpoints, one boundary.

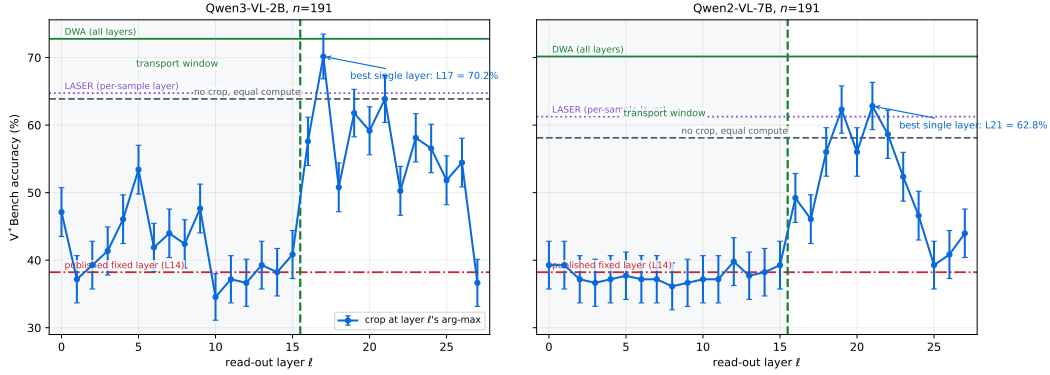


Figure 2: **End-task accuracy of cropping at each layer’s arg-max, both checkpoints.** Every arm answers at ~ 600 visual tokens, so only the read-out depth varies. The spread from depth alone is 35.6 points on Qwen3 and 26.7 on Qwen2, and inside the transport window no layer beats simply not cropping on either. The curve steps up at the boundary and peaks at L17 (Qwen3) / L21 (Qwen2). DWA reads all 28 layers and sits above every single layer on both: against the best one — chosen on the test set, so an optimistic baseline — it is +2.6 n.s. on Qwen3 and +7.3 [+1.0, +13.6] on Qwen2, i.e. it matches or beats an oracle choice of layer without being told which layer that is. The published fixed choice (L14) is 24–32 points below that oracle.

The consequence. The read-out band used by every method in this family (L17–20 on Qwen3, L19–22 on Qwen2) lies *entirely after* the transport window. Those layers’ image attention is causally inert ($KL \approx 0.001$). This predicts, and we confirm, three things: depth *re-weighting* helps while depth *selection* never does (§6.1); tokens can be dropped at the boundary for free (§6.2); and placement rules that all read the same inert depth separate only modestly.

Selection stops mattering exactly at the boundary. Before it, the choice of which tokens to keep is worth +8.4 and +8.9 points over random. At and after it, ranking is worth nothing: random keep equals attention keep on both models at every keep ratio. The boundary is not merely where pruning becomes cheap — it is where selection ceases to matter.

6 TWO POLICIES THAT FOLLOW FROM THE MEASUREMENT

Both policies below are derived from the boundary located in §5 rather than tuned: one replaces the hand-picked read-out depth, the other exploits the fact that visual tokens are causally inert past the boundary. Neither introduces a hyperparameter chosen on test data.

6.1 DEPTH-WEIGHTED ATTENTION (DWA)

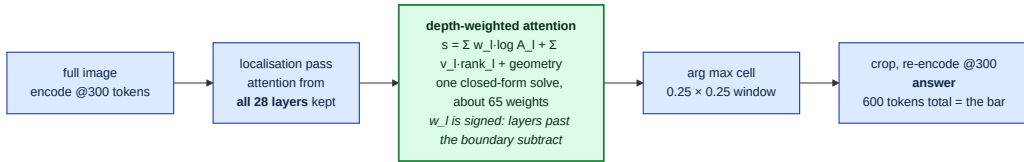


Figure 3: **Policy I: Depth-Weighted Attention (DWA).** One localisation pass, every layer’s attention kept, a single closed-form solve that re-weights depths, and a crop. All 28 layers enter the score. The fitted weights are signed, but sign is not the mechanism: a non-negative refit retains the whole advantage while zeroing 44 of 56 per-layer coefficients (§J).

Localise at 300 tokens with the full prompt (the read-out must be taken at the answer-emission position; without the answer instruction the map loses 36–42 points). Read the final prompt token’s

attention over image cells at all 28 layers. Score each cell by a single closed-form ridge solve,

$$s(\text{cell}) = \sum_{\ell} w_{\ell} \log A_{\ell} + \sum_{\ell} v_{\ell} \text{rank}_{\ell} + a \log \text{nb} + \text{geometry},$$

fit on as few as 50 boxed examples. Crop $W=0.25$ at the arg-max, re-encode at 300 tokens, answer.

The fitted w_{ℓ} is the depth filter (Figure 1). It takes both signs: it assigns *negative* weight to 7 of the 11 layers the conventional block-mean read-out adds at +1, and its largest positive weight falls in the band where attention first becomes question-conditioned. There is no boosting, no learning rate, no early stopping, no seed averaging — given the data the solution is unique.

Why re-weighting and not selection. On the items the conventional read-out gets wrong, the learned head ranks the target cell above the cell the arg-max chose on 57% and 71% of items (gradient-boosted; the deployed ridge ties it as a ranker, Appendix E), against 13% and 12% for the block mean. None of the simple summaries of the depth profile reproduces this — not magnitude, neighbourhood, isolation, or a windowed sum. The mechanism is a *supervised depth weighting*, not subtraction: the block mean’s failure is caused by the item-independent component of the maps, and a non-negative fit — which zeroes 44 of 56 per-layer coefficients — retains the full advantage on both checkpoints and matches DWA at the end task (Appendix J).

Ablations. A gradient-boosted tree over the same features is statistically indistinguishable from the ridge solve on all four architectures, so the closed form costs nothing. Capacity is not the limit at this sample size: a spatial CNN is *significantly worse* on the 7B (−8.9 and −9.4 at two window sizes), and nine random seeds, flip augmentation and ensembling all land inside the noise floor. Roughly 50 boxed examples recover 70% of the full-data gain. A training-free variant — the maximum over the four question-conditioned layers — reaches 68.9 and 60.7: above every published rule on Qwen3, and above the block-mean and fixed-layer rules on Qwen2 (LASER reaches 61.3 there), but 3.7 and 9.5 points below the fitted filter.

6.2 ADAPTIVE VISUAL RESOLUTION (AVR)

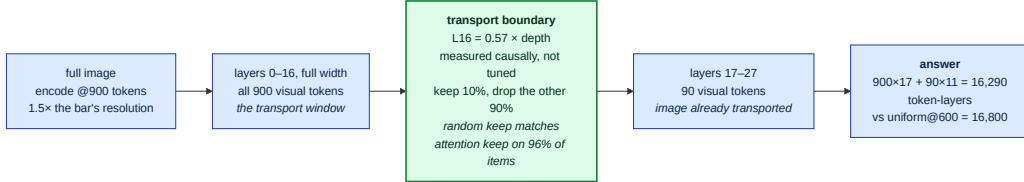


Figure 4: **Policy II, adaptive visual resolution.** Spend the compute the inert late-layer visual tokens would have used on resolution inside the window where the image is actually read. The prune depth is read off the causal measurement rather than tuned, and nothing spatial is discarded before the boundary.

Because visual tokens are inert past the boundary, discard them there and spend the saving on pixels: encode at 900 tokens, run full width through L0–L16, keep the top 10% at L16, run L17–27 narrow. Token-layers $900 \times 17 + 90 \times 11 = 16,290$ — 96% of the bar, for $1.5 \times$ the resolution. No crop.

Pruning is free to within ± 0.5 points in 7 of the 10 measured cells (two more are ahead by 1.3 and 2.6, read as noise), at every keep ratio (10/25/50%), with any ranking including random; one cell shows a real cost (HR-8K cross-instance, −1.8). Output probabilities change on 99% of items while the arg-max changes on 2% — the tokens are inert, not absent.

An identity. Across 4 checkpoints $\times$ 3 benchmarks, AVR’s gain equals the checkpoint’s *measurable resolution headroom* $\text{acc}(900) - \text{acc}(600)$ on that stratum: $r = 0.966$, slope 1.01, mean absolute error 0.7 points over 10 stratum-cells. The headroom needs two baseline runs and no method at all, so AVR’s benefit is *predictable before deployment* — and it explains the cells where AVR gains little: those are the cells where the checkpoint gains little from resolution in the first place, including one where the baseline itself degrades from 600 to 900 tokens.

The boundary is the right depth, not merely a cheap one. One cut at the boundary dominates the schedules we tried: a taper that prunes inside the transport window loses on both checkpoints (-1.0 Qwen3, -5.8 Qwen2), and on the 36-layer checkpoint a voided run that cut at 0.14 of depth instead of 0.57 cost 7.9 points pooled (Appendix M). The FastV default (layer 2) is the exception that proves the point: it costs 4.7 on Qwen3 but is the better schedule on Qwen2, whose resolution curve is flat beyond 600 tokens — the best fixed schedule differs by checkpoint, so we claim only that the cut must not fall inside the window. Pruning *after* the boundary saves less for no additional accuracy.

7 EVALUATION

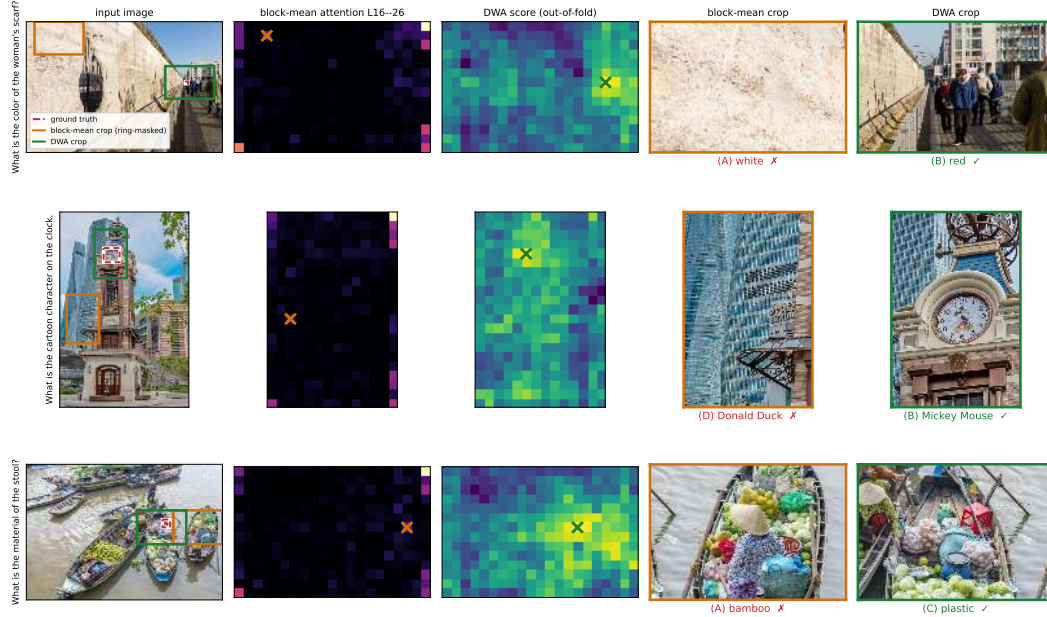


Figure 5: **What the two read-out rules see, on real V*Bench items** (Qwen3-VL-2B). The block-mean arg-max is shown as the paper evaluates it, which masks the grid’s outer ring before taking the arg-max — and that mask is doing very unequal amounts of work. Without it, the block-mean arg-max lands in the grid’s *last column*, the position before a raster row-wrap, on 84.8% and 61.8% of items across the two checkpoints against a 5.0% chance rate, and covers half the ground-truth box on only 13.6% of items; the ring mask lifts that to 46.6% and 39.3% across the two checkpoints. DWA covers 63.9% and 55.0%, and barely uses the mask: it is worth $+0.012$ and $+0.003$ mean coverage to DWA against $+0.336$ and $+0.079$ to block-mean, an order of magnitude on both. DWA removes the serialisation sink by a supervised depth weighting that suppresses the item-independent component, not by patching the geometry and not by the signs of the fitted weights (Appendix J). *Examples are not hand-picked*: we take single-instance items where the DWA crop covers the box and the block-mean crop does not *and* the answers flip accordingly, restricted to items where the recomputed out-of-fold DWA cell reproduces the logged run exactly. Nine items qualify; these are the first three by question id.

DWA beats every published rule on every cell where that rule was run — 9 of 9 paired contrasts, all CIs clear: $+3.8$ to $+15.9$ against ViCrop-style, $+7.9$ and $+8.9$ against LASER, $+31.9$ and $+34.6$ against fixed-layer. At matched compute the published rules are worth -1.6 to $+3.1$ against simply spending the tokens on resolution.

7.1 NATIVE DYNAMIC RESOLUTION

The field publishes these benchmarks at the model’s native dynamic resolution. On V*Bench that is a median of 3,290 visual tokens — $5.5\times$ our bar — and 76.3% accuracy.

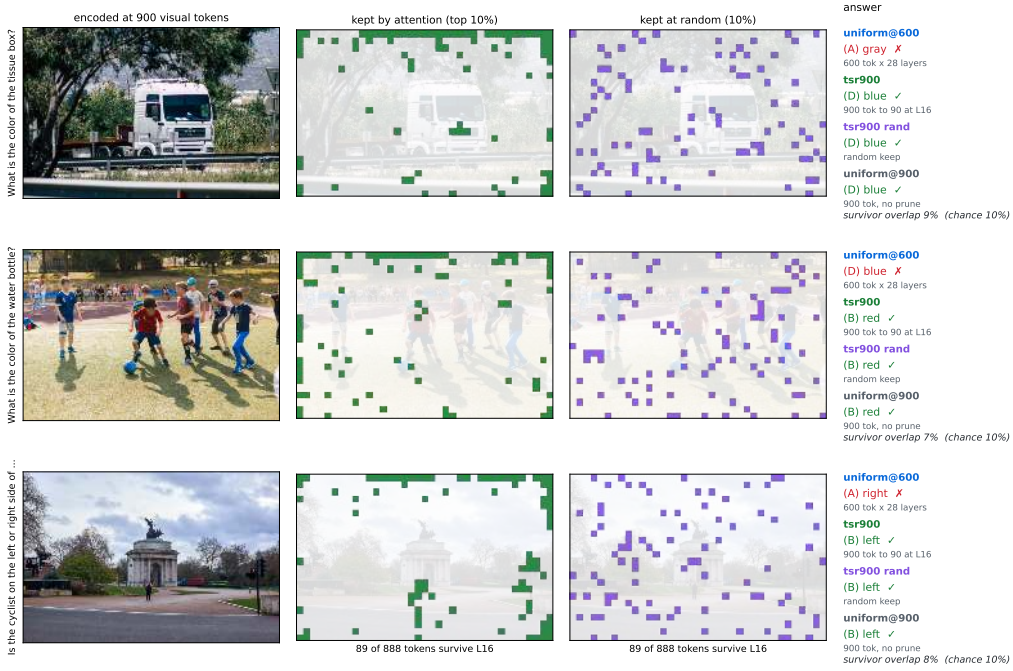


Figure 6: **What survives the transport boundary, and why the choice of survivor does not matter** (Qwen3-VL-2B). Of 888 visual tokens, 89 are carried past L16. The target encode is 900 tokens; the processor’s own grid yields 888 on these images, inside the $\pm 8\%$ tolerance of Appendix A, and the arithmetic quoted elsewhere ($900 \times 17 + 90 \times 11$) is at the target. Ranking them by attention selects mostly border tokens — the same sink Figure 5 exposes — while a random keep of the same size overlaps it by only 7–9%, at the 10% chance rate. Both give the same answer, and so does not pruning at all, on all three items; across the benchmark attention-keep and random-keep agree on 96.3% of items (70.2% vs 68.6% accuracy). The gain over the bar is resolution, not selection. Selection rule for the examples: AVR correct, the equal-compute bar wrong, and random keep agreeing with attention keep; 15 items qualify.

Table 2: All placement rules in one run, V*Bench, every arm at ~ 600 visual tokens. $\checkmark$ = 95% CI clear. $n=190$ on Qwen3 and $n=191$ on Qwen2: one item lacks the LASER arm on Qwen3 and is dropped from every arm so all rules are compared on identical items. Table 3 uses the same item sets.

method	Qwen3-VL-2B			Qwen2-VL-7B		
	single	cross	ALL	single	cross	ALL
uniform@600 (bar)	62.3	65.8	63.7	57.4	59.2	58.1
fixed layer L14	24.6	57.9	37.9	33.9	44.7	38.2
block-mean arg-max (ViCrop-style)	64.0	59.2	62.1	67.0	47.4	59.2
LASER (Zhu et al., 2026)	64.0	65.8	64.7	65.2	55.3	61.3
gate-max (ours, no training)	72.8	63.2	68.9	63.5	56.6	60.7
DWA (ours)	77.2	65.8	72.6	71.3	68.4	70.2
oracle crop (upper bound)	96.5	75.0	87.9	93.9	77.6	87.4

Three readings, both checkpoints. (i) DWA is *statistically indistinguishable* from native dynamic resolution (-3.7 and -1.0 , CIs spanning zero) while using 18% and 14% of native’s visual-token compute — whereas every published rule is *significantly worse* than native (-9.9 to -14.2). The separation between our read-out and theirs is therefore not an artefact of the imposed budget; it is larger in the regime the field actually publishes in. (ii) Per stratum on Qwen3, DWA at 600 tokens matches native on single-instance ($+2.6$) and AVR matches it on cross-instance (-2.6), each at a fifth of the compute. (iii) A perfectly placed 600-token crop *beats* native by $+11.6$ and $+16.2$

Table 3: **Against native dynamic resolution.** V*Bench. Our arms use 600 visual tokens; native uses 3,290 (Qwen3) and 4,320 (Qwen2), i.e. $5.5\times$ and $7.2\times$ more. ✓/✗ mark CIs clear of zero.

arm	Qwen3-VL-2B ($n=190$)		Qwen2-VL-7B ($n=191$)	
	accuracy	vs. NATIVE	accuracy	vs. NATIVE
NATIVE dynamic resolution	76.3	—	71.2	—
uniform@600 (bar)	63.7	−12.6 ✗	58.1	−13.1 ✗
block-mean arg-max	62.1	−14.2 ✗	59.2	−12.0 ✗
LASER (Zhu et al., 2026)	64.7	−11.6 ✗	61.3	−9.9 ✗
AVR (ours)	70.0	−6.3	61.8	−9.4 ✗
DWA (ours)	72.6	−3.7	70.2	−1.0
<i>oracle crop @600</i>	<i>87.9</i>	<i>+11.6 ✓</i>	<i>87.4</i>	<i>+16.2 ✓</i>

pooled, and by +21.7 on single-instance (Qwen3): **placement is worth more than 5–7× the compute**. This is the strongest single argument for allocation research, and it is a ceiling nobody has approached.

7.2 FOUR ARCHITECTURES, AND WHAT EQUAL COMPUTE DOES TO THE COMPARISON

The results so far use two Qwen checkpoints, whose resolution axis is continuous: a crop can be encoded at exactly half the bar, so *localise@300* + *crop@300* is budget-matched against *uniform@600* by construction. Most other families quantise resolution — anyres tilers and fixed-token encoders cannot reach half the bar at all — so a two-pass method there necessarily costs more than one pass, and the comparison has to be made explicitly rather than assumed. We therefore report, for every family, the visual-token cost of each arm and a *bar_matched* arm: a *single* pass given the same total budget the two-pass method consumes.

Table 4: **DWA across four architectures**, V*Bench, $n=191$ each. *block* is the published single-block read-out, the incumbent. *cost* is DWA’s total visual tokens over the bar’s. ✓ = 95% CI excludes zero. **DWA – block** is cost-fair: both arms pay one localise pass and one crop at the same budget. **DWA – bar_matched** asks the different question of whether cropping beats spending the same compute on resolution.

model	bar	block	DWA	bar_m	cost	DWA–block	DWA–bar_m
Qwen3-VL-2B	63.9	62.3	72.8	—	1.00×	+10.5 ✓	+8.9 ✓
Qwen2-VL-7B	58.1	59.2	70.2	—	1.00×	+11.0 ✓	+12.0 ✓
Gemma-3-4B	36.1	36.6	52.9	35.6	2.00×	+16.2 ✓	+17.3 ✓
InternVL3-8B	60.2	66.0	76.4	70.7	8.37×	+10.5 ✓	+5.8
LLaVA-OneVision-7B	52.9	61.8	67.0	66.0	2.27×	+5.2	+1.0

The read-out claim holds on four of five families. Depth re-weighting beats the published single-block read-out at matched cost on Qwen3, Qwen2, Gemma-3 and InternVL3, and is positive but not significant on LLaVA-OneVision. The families span four different visual-token designs: native dynamic resolution, a fixed 256-token encoder, 448-pixel dynamic tiling, and anyres.

Gemma-3 is the cleanest case, because there the incumbent does nothing at all. Its block-mean arg-max covers ground-truth boxes at 0.071 against a 0.066 chance rate — $1.1\times$, i.e. indistinguishable from random — and is worth +0.5 points [−6.8, +7.9] at the end task. The depth-weighted read-out of *the same attention* reaches 0.395 coverage ($6.0\times$ chance) and +16.2 points. Selecting one depth band recovers nothing; re-weighting depth recovers sixteen points.

On the stronger claim we are narrower, and the headroom identity says why. Against a bar given the same total budget, DWA wins on three of five. It fails on exactly the two families that still have unspent resolution headroom: LLaVA-OneVision gains 13.1 points from 52.9 to 66.0 simply by encoding at the higher budget, and InternVL3 gains 10.5. Gemma-3, whose encoder emits 256 tokens for every image, gains −0.5: it *cannot* convert compute into resolution, and there DWA’s advantage is largest. This is the same identity that governs AVR (Appendix F) now governing when

placement is worth buying at all. Qwen is the exception that keeps it a regularity rather than a law: it has headroom and DWA still wins.

A caution about how this family is usually evaluated. The column the literature reports is DWA — bar, and it is systematically the most flattering one: +14.1 ✓ on LLaVA-OneVision and +16.2 ✓ on InternVL3, both of which fall to a non-significant +1.0 and +5.8 once the bar is given the same compute. InternVL3’s $8.37\times$ ratio is itself an artefact worth naming: its ladder jumps from 256 tokens to 1792, so a 600-token target lands on 256 while the crop costs 1792. We report the matched column throughout and recommend it be required.

7.3 A SCOPE LAW

Table 5: **The scope law.** Accuracy difference vs. the equal-compute bar, per stratum, all seven benchmark $\times$ checkpoint cells. Cropping helps single-instance everywhere and is non-positive on cross-instance almost everywhere; AVR is positive on both strata in 6 of 7 cells but gives up single-instance gain.

cell	DWA (2 passes, ~ 50 boxes)		AVR (1 pass, no supervision)	
	single	cross	single	cross
V*Bench Qwen3-VL-2B	+14.8 ✓	+0.0	+3.5	+10.5 ✓
V*Bench Qwen2-VL-7B	+13.9 ✓	+9.2	+3.5	+1.3
V*Bench Qwen2.5-VL-7B	+12.2 ✓	−2.6	+12.2 ✓	+7.9
V*Bench Qwen3-VL-8B	+6.1	−3.9	+7.0	+7.9 ✓
HR-4K Qwen3-VL-2B	+10.2 ✓	−8.8 ×	+4.0 ✓	−2.8
HR-4K Qwen2-VL-7B	+9.2 ✓	−6.0 ×	+2.8	+0.8
HR-8K Qwen3-VL-2B	+8.8 ✓	−8.8 ×	+6.5 ✓	+1.8
positive cells	7/7	1/7	7/7	6/7
significant losses	0	3	0	0

Across 14 cells (3 benchmarks $\times$ 4 checkpoints $\times$ 2 strata), attention-guided cropping helps single-instance perception in 7/7 and is non-positive on cross-instance in 6/7, significantly negative in three (−6.0, −8.8, −8.8). This holds for *every* method in the family, ours included. The cause is partly, but only partly, geometric. On V*Bench’s 54 multi-box items (mean 1.71 boxes) the union of the boxes fits inside a single $W=0.25$ window on 59.3% of them: for the other $\sim 41\%$ no fixed window of that size can contain the evidence, so the fixed window is itself a cause and not merely the placement. On the 59.3% where a containing window does exist, the relation is still not recovered from the magnified view (Appendix L) — placement is not the binding constraint there. AVR shows the opposite profile — positive on both strata in 6/7 cells with no significant loss anywhere — but gives up 3–10 points of single-instance gain.

The ceiling explains why cropping cannot be repaired here. With *perfect* placement, cross-instance still fails 25% and 22% of the time against 3.5% and 6.1% for single-instance. Cross-instance error is not predominantly perceptual, and no crop, multi-crop, composition or pruning schedule reached it (§8).

7.4 PRACTICAL GUIDANCE

The two policies have different cost and risk profiles, and the choice does not require a question classifier. The DWA needs ~ 50 boxed examples, two forward passes and $1.4\text{--}2.0\times$ wall-clock; it has the higher ceiling (+14.8 on single-instance) and a 6–9 point cross-instance failure mode. AVR needs no supervision, one forward pass and $\sim 1\times$ wall-clock; it is positive on both strata in 6 of 7 cells with no significant loss anywhere, and its benefit is predictable from two baseline runs. Head to head the two are even pooled (mean difference +0.2 points over seven cells). We recommend AVR as the default and DWA where boxed supervision exists and single-instance perception dominates. We note that DWA’s pooled margin over AVR is concentrated on the two cells on which it was developed.

8 WHAT DOES NOT WORK

Ten designs failed to make cropping serve cross-instance questions, and two more failed to combine the two policies into a single pipeline. We report them because the diagnoses constrain the design space more than the successes do. Each failed for a measured reason: per-item window sizing; peak-spanning and mass-containment windows; joint (cell, window) re-ranking under a coverage objective (a metric artefact — coverage is monotone in window size); expected-accuracy window selection; a downscaled scene composed with the crop, which dilutes it by 8.7 and 5.2 points *even at oracle placement*; multi-crop at matched budget, where the second crop genuinely converts (+7.8 ✓) but halving each crop costs −11.3; and the same at full resolution funded by pruning. Depth *selection* rules ported from the LLM literature — DoLa, SLED, confident-layer decoding, attention-JSD, rank-stability — all fail, for one reason: in these models final-layer entropy rises on 54–88% of items while per-layer accuracy still climbs monotonically to the last layer. There is no better layer to select.

9 LIMITATIONS

$n = 191$ on V*Bench gives per-stratum CI half-widths near 10 points; no method is significantly positive on both strata in any single cell, which is a power limitation as much as a method one. DWA needs ~ 50 boxed examples and two forward passes; AVR needs neither. DWA’s pooled margin over AVR is concentrated on the two cells it was developed on. Pruning at the boundary is free in 9 of 10 cells; the exception (HR-Bench 8K, cross-instance, $-1.8 \times$) is real.

The two claims have different scope, and we do not merge them. That depth re-weighting beats the published single-block read-out holds at matched cost on four of five architectures (Table 4); on LLaVA-OneVision it is positive but not significant. That cropping is worth its compute *at all*, against a bar given the same total budget, holds on only three of five, failing on the two families with unspent resolution headroom. A practitioner whose checkpoint can still buy resolution should buy resolution first.

The transfer arm is adopted on a threshold it clears exactly. Its pre-registered rule — [DWA — block] excluding zero on ≥ 2 of 4 Qwen cells — passes 2 of 4, and the split is not random: both Qwen2-VL-7B cells pass and both Qwen3-VL-2B cells fail. Stratified, the single-instance margin is the consistent quantity (+9.5, +5.0, +10.5, +10.0; three of four CIs clear).

Two of the five families required a measured token layout before any method could be run at all, and a wrong layout is empirically indistinguishable from an incompatible architecture: our first attempt put Gemma-3 and LLaVA-NeXT at $1.1\times$ and $1.14\times$ chance and we wrongly concluded both were incompatible. Attention-based allocation is more sensitive to visual-token layout than this literature acknowledges, and we report the layout we measured for each family rather than deriving it from tile arithmetic (Appendix M).

10 CONCLUSION

Attention-guided cropping reads the model’s attention at a depth where that attention no longer carries the image to the answer. Measuring where it does — a causal masking experiment, not a heuristic — yields both a better read-out (DWA) and a free pruning point (AVR), and the two together match a model running at $5.5\times$ the visual-token budget. The remaining gap is placement: perfect placement at 600 tokens exceeds native dynamic resolution by 11.6 points.

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A EXPERIMENTAL PROTOCOL

Models. Qwen3-VL-2B-Instruct and Qwen2-VL-7B-Instruct (core pair), Qwen2.5-VL-7B-Instruct, Qwen3-VL-8B-Instruct (36 layers), LLaVA-NeXT-7B and LLaVA-OneVision-7B (attention analyses only). `bfloat16`; eager attention wherever hooks or a prune bias are required, SDPA otherwise. We note a ~ 2 point kernel-dependent difference on the 7B between the two implementations and always compare arms computed under the same kernel.

Budgets. Images are resized so that the processor’s own `image_grid_thw` yields the target token count ($\pm 8\%$); realised counts are recorded per arm and a contrast is voided if budgets drift by more than 10%. Where pruning changes cost mid-network we report token-layers.

Read-out. The attention map is the final prompt token’s attention over image-token columns, averaged over heads *before* normalising (the order matters: normalising per head first up-weights sink-heavy heads and shifts the block-mean read-out by 11 points on the 7B). The prompt must end at the answer-emission point; a prompt ending elsewhere costs 36–42 points on the arg-max read-out.

Statistics. Paired item bootstrap with 8000 resamples. Learned components use GroupKFold(5) grouped by item $\times$ 3 seeds; the headline DWA result is additionally reproduced under a disjoint fold-seed set (difference within ± 1 point). Run-to-run floor on coverage is 1–2.5 points.

Pipeline discipline. An exactly +0.0 contrast with a $[+0.0, +0.0]$ interval is treated as a pipeline fault, never a result; this rule caught a monkey-patch that silently failed to reach one model family. Every pruning run asserts at runtime that the loaded model’s attention module is the one that was patched.

B INTERMEDIATE METRICS SYSTEMATICALLY OVER-REPORT

Five times in this work an intermediate metric improved while end-task accuracy did not: a read-out that closed the coverage gap but converted 3.1 points worse; a ridge variant that tied on coverage and differed on the chosen cell for two-thirds of items; an early-exit localiser free in coverage but -3.7 pooled at the end task; a higher-resolution localiser that gained $+4.2$ coverage and -3.7 accuracy; and an objectness probe that retains the evidence box far better than attention (87% vs. 72%) and converts to -1.6 . We therefore report coverage only as a diagnostic and gate every claim on end-task accuracy at matched compute.

C THE TRANSPORT MEASUREMENT IN FULL

A forward pre-hook on every decoder layer adds -10^4 to the attention logits of chosen query rows over the image-token columns, at chosen layers only. Masking *all* text rows at *all* layers destroys the answer (KL 1.266 / 0.532, 42% / 45% of answers flip, $n=40$); masking only the answer token’s row at all layers does not (KL 0.025, 3% flip). Table 6 localises the window.

Table 6: Prefix and suffix masks, all text rows, $n = 40$ per model, V*Bench at 300 tokens. The prefix mask $L_{0..\ell}$ saturates at the full-mask value once ℓ passes the window; the suffix mask $L_{\ell..end}$ falls to zero once ℓ is past it. Both read the boundary at the same place, and the 36-layer checkpoint puts it at the same fraction of depth ($\ell=20$ of 36).

	$\ell=4$	$\ell=8$	$\ell=12$	$\ell=16$	$\ell=20$	$\ell=24$
Qwen3 prefix $L_{0..\ell}$ (KL)	0.082	0.549	1.278	1.259	1.215	1.257
Qwen3 suffix $L_{\ell..}$ (KL)	0.967	0.880	0.320	0.014	0.002	0.001
Qwen3 suffix (answers flipped)	53%	50%	20%	12%	0%	0%
Qwen2 prefix $L_{0..\ell}$ (KL)	0.037	0.105	0.364	0.547	0.534	0.538
Qwen2 suffix $L_{\ell..}$ (KL)	0.518	0.465	0.429	0.029	0.004	0.002
Qwen2 suffix (answers flipped)	53%	47%	40%	5%	3%	0%
Qwen3-8B (36L) prefix $L_{0..\ell}$ (KL)	0.100	0.309	1.190	1.740	1.559	1.565
Qwen3-8B (36L) suffix $L_{\ell..}$ (KL)	1.671	1.538	1.278	0.413	0.052	0.009
Qwen3-8B (36L) suffix (answers flipped)	35%	45%	40%	15%	5%	0%

Single-layer masks peak at L11 on Qwen3 (KL 0.504; next highest L13 at 0.101) and at L15/L14 on Qwen2 (0.118/0.096). From L20 onward every layer is at or below 0.0023 on both models. On Qwen3-VL-8B, which has 36 layers, the same behaviour appears at L21, i.e. the same 0.57 fraction of depth.

Three independent readings of the same boundary. The masks are one. Pruning is the second: discarding 90% of visual tokens at the boundary costs +0.0 to +0.5 points across ten stratum-cells, whereas moving the same cut to $\ell=4$ costs 5.2 more on one model (0.5 on the other), and a taper that removes half the tokens at $\ell=9$ –12 loses 3.5 points against not pruning at matched tokens while buying nothing over a single cut (-1.0). Read-out depth is the third: restricting the localiser to

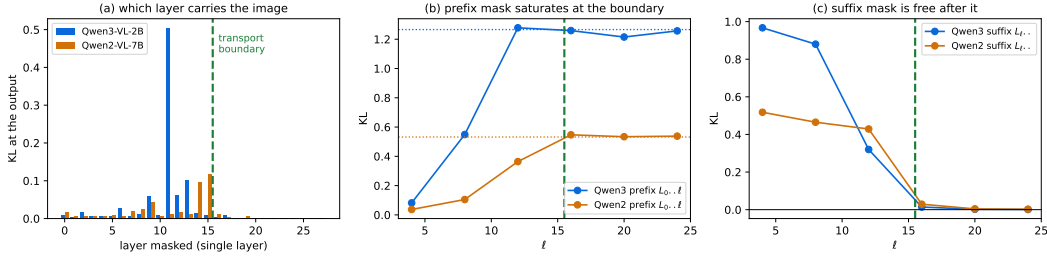


Figure 7: **The transport measurement.** (a) Masking one layer’s image attention at a time: the effect is concentrated well before the boundary and is at or below 0.0023 from L20 on both models. (b) Masking a prefix $L_{0..l}$ reaches the full-mask value (dotted) once l passes the window. (c) Masking a suffix $L_{l..end}$ costs nothing once l is past it. Three readings, one boundary.

layers $\leq$ L20 costs -0.5 and $+1.6$ points of coverage, while restricting it to $\leq$ L16 costs 9.4 and 22.0. Transport finishes at the boundary; the signal the ranking reads is produced after it.

D DEPTH STRUCTURE ACROSS CHECKPOINTS

Table 7: Where each quantity falls, per checkpoint. The transport boundary and the read-out band are fractions of depth, not fixed layer indices, which is what lets both policies transfer to a 36-layer model unchanged. The 36-layer boundary is masked: its suffix is harmless at L20 and zero from L24, the same fractions as the pair.

checkpoint	layers	block-mean band	read-out band	boundary	boundary / depth
Qwen3-VL-2B	28	L16–L26	L17–L20	L16	0.57
Qwen2-VL-7B	28	L15–L26	L19–L22	L16	0.57
Qwen2.5-VL-7B	28	L16–L26	—	L16	0.57
Qwen3-VL-8B	36	L20–L33	—	L21	0.57

E FULL PAIRED COMPARISON AGAINST PUBLISHED RULES

Table 8: DWA minus each alternative on V*Bench, paired on identical items, pooled over strata, $n = 191$ (190 for LASER, one item missing). All six published-rule contrasts clear zero on both models. Bootstrap 95% CIs, 8,000 resamples, matching the protocol in Appendix A.

contrast	Qwen3-VL-2B	Qwen2-VL-7B
– uniform@600 (equal-compute bar)	+8.9 [+0.5, +17.3] ✓	+12.0 [+4.7, +19.4] ✓
– fixed-layer crop (L14)	+34.6 [+25.7, +43.5] ✓	+31.9 [+24.1, +39.8] ✓
– block-mean arg-max	+10.5 [+4.2, +16.8] ✓	+11.0 [+4.7, +17.8] ✓
– LASER	+7.9 [+1.1, +14.7] ✓	+8.9 [+3.1, +14.7] ✓
<i>controls, not baselines</i>		
– gate-max	+3.7 [−2.1, +9.4]	+9.4 [+3.7, +15.2] ✓
– gradient-boosted tree	−0.5 [−4.7, +4.2]	+7.3 [+2.1, +13.1] ✓
– DWA refit, disjoint fold seeds	−1.0 [−3.1, +1.0]	+1.0 [−1.0, +3.1]
– oracle placement	−15.2 [−22.5, −8.4] ✗	−17.3 [−24.6, −9.9] ✗

Taking these six together with the three pooled block-mean contrasts in Table 9, DWA beats a published rule in nine of nine pooled comparisons. It does not beat the equal-compute bar everywhere, which is a separate claim and the subject of the scope law. The fold-seed control matters most. Refitting DWA under three disjoint fold seeds moves the pooled result by 1.0 point or less on both models, so the margin is not an artefact of which items landed in which fold. The oracle row is the remaining headroom: even the best rule we have is 15–17 points below perfect placement.

Table 9: HR-Bench, DWA minus each alternative, $n = 400$ per stratum, 800 per cell. DWA is transferred: its weights were fit on V*Bench and applied here with nothing refit. Placement quality replicates; the scope law replicates with it.

		HR-4K Qwen3-2B	HR-4K Qwen2-7B	HR-8K Qwen3-2B
vs. bar	single	+10.2 [+5.5, +15.0] ✓	+9.2 [+4.5, +14.2] ✓	+8.8 [+3.5, +13.8] ✓
	cross	−8.8 [−14.0, −3.5] ✗	−6.0 [−11.5, −0.8] ✗	−8.8 [−14.5, −2.8] ✗
	pooled	+0.8 [−2.8, +4.4]	+1.6 [−1.9, +5.2]	+0.0 [−3.9, +4.0]
vs. block-mean	single	+7.8 [+4.0, +11.5] ✓	+15.0 [+10.2, +19.8] ✓	+5.2 [+1.2, +9.2] ✓
	cross	+1.8 [−2.8, +6.2]	+16.8 [+12.0, +21.5] ✓	+2.2 [−2.5, +6.8]
	pooled	+4.8 [+1.9, +7.6] ✓	+15.9 [+12.6, +19.2] ✓	+3.8 [+0.6, +6.9] ✓
vs. random cell	single	+32.8 [+26.5, +39.0] ✓	+24.8 [+18.8, +30.5] ✓	+24.2 [+18.0, +30.2] ✓
	cross	+10.8 [+5.0, +16.5] ✓	+15.8 [+10.5, +21.0] ✓	+7.0 [+2.0, +12.0] ✓
	pooled	+21.8 [+17.5, +26.1] ✓	+20.2 [+16.4, +24.2] ✓	+15.6 [+11.6, +19.6] ✓

The pooled rows against the bar are the scope law in its plainest form. Cropping wins 9–10 points where the question is about one instance, loses 6–9 where it is about a relation between two, and the two cancel to nothing in aggregate. A paper that reported only the pooled column would conclude the method does not work; a paper that reported only the single-instance column would conclude it works everywhere. Neither is true.

F THE RESOLUTION-HEADROOM IDENTITY

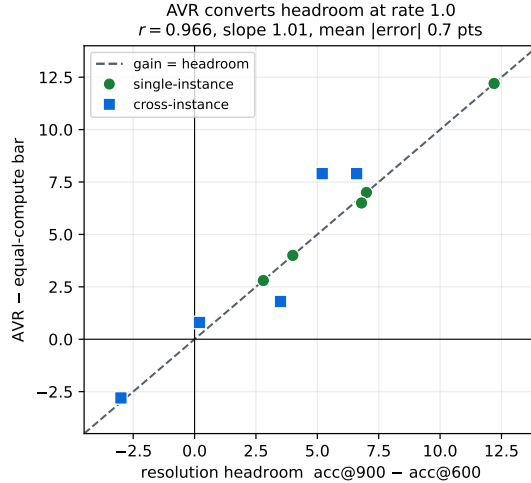


Figure 8: **AVR is a resolution converter.** Its gain over the equal-compute bar tracks the headroom the checkpoint already has, at rate 1.0. The one cell where AVR loses is the one cell where the baseline itself is worse at 900 tokens than at 600.

The identity also holds where it predicts *nothing*. Applying AVR to a crop rather than to the full image gives eight further cells whose headroom is near zero; gain tracks headroom there with $r = 0.911$ and slope 1.03 (mean error 0.40 points), and AVR correctly does nothing. Across both settings the identity is supported on 18 stratum-cells, and it is the only component of this work that predicted a negative result before we ran it.

Two consequences. First, the sign of the gain is predictable before deployment: the one cell where AVR loses (HR-4K cross-instance) is the one cell where the baseline itself is worse at 900 tokens than at 600, so nothing was available to convert. Second, the pruning contributes nothing on its own. Comparing AVR against `uniform@900`, which encodes at the same resolution and prunes nothing, the two agree within 0.5 points in 7 of the 10 cells (+0.0, +0.0, +0.0, +0.0, +0.2, +0.5,

Table 10: AVR’s gain against the headroom $\text{acc}(900) - \text{acc}(600)$, for the ten stratum-cells that carry the `uniform@900` diagnostic. Pearson $r = 0.966$, OLS slope 1.01, mean $|\text{gain} - \text{headroom}| = 0.7$ points. Headroom is measurable from two baseline runs with no method involved.

cell	stratum	AVR – bar	headroom
V*Bench Qwen2.5-VL-7B	single	+12.2 ✓	+12.2
	cross	+7.9	+5.2
V*Bench Qwen3-VL-8B	single	+7.0	+7.0
	cross	+7.9 ✓	+6.6
HR-4K Qwen3-VL-2B	single	+4.0 ✓	+4.0
	cross	−2.8	−3.0
HR-4K Qwen2-VL-7B	single	+2.8	+2.8
	cross	+0.8	+0.2
HR-8K Qwen3-VL-2B	single	+6.5 ✓	+6.8
	cross	+1.8	+3.5

−0.2). In two more the pruned arm is *ahead*, by +1.3 and +2.6, which we read as noise rather than a benefit of pruning. Exactly one cell shows a real cost: HR-8K cross-instance at −1.8 [−3.2, −0.5], the only measurable price of discarding 90% of visual tokens anywhere in this work. Pruning at the boundary is therefore free everywhere we have measured it but that one cell. We verified the pruning is applied rather than silently skipped at the distribution level: output probabilities change on 99% of items while the arg-max changes on 2%. The discarded tokens are inert, not absent.

G CLAIM-TO-EVIDENCE MAP

Every headline claim, its strongest evidence, and its model count. *Intervention* means the claim was tested by changing the model’s computation (mask, patch, prune, ablate, oracle), not by observation alone. A claim is exported only if it holds on two checkpoints.

Table 11: Claims, evidence class, and the models on which the evidence was measured.

claim	strongest evidence	class	models
transport complete by 0.57× depth	layer-wise and prefix/suffix attention masks; boundary from pruning cost	intervention	2 × 28-layer masking; 36-layer pruning
image-token values inert past the boundary	representation patching at $L \geq 16$, at 300 and 900 tokens	intervention	2 models, both strata
answer position does not read the image	answer-row-only mask, all layers	intervention	2 models
depth choice dominates usable read-out band lies after transport	crop at every layer’s arg-max, end task coverage curve, end-task sweep, location probe	intervention measurement	2 models + 2 models (curve, probe)
ranking matters before the boundary only	probe vs random keep at L4; random = attention at L16/L24; value patching	intervention	2 models
block mean fails on the item-independent component	nuisance ablation: 0.083→0.438, 0.292→0.490	intervention	2 models
supervised depth weighting, not signed subtraction	non-negative fit retains the advantage (44/43 of 56 coefficients zeroed); end task n.s.	intervention	2 models
DWA beats every published rule at equal compute	9/9 pooled paired contrasts	evaluation	2 models × 2 bench-marks
AVR pruning is free	pruned vs unpruned @900; value patching at L16	intervention	2 models
AVR’s gain is resolution headroom	10-cell identity, $r = 0.966$; distributional equivalence	accounting + intervention	4 checkpoints
cropping helps single-instance, not cross	14-cell scope table + oracle ceiling	evaluation + intervention	4 checkpoints
DWA matches native resolution at 14–18% compute	end-task vs native	evaluation	2 models
placement is worth > 5× the compute	oracle-crop arm beats native	intervention	2 models

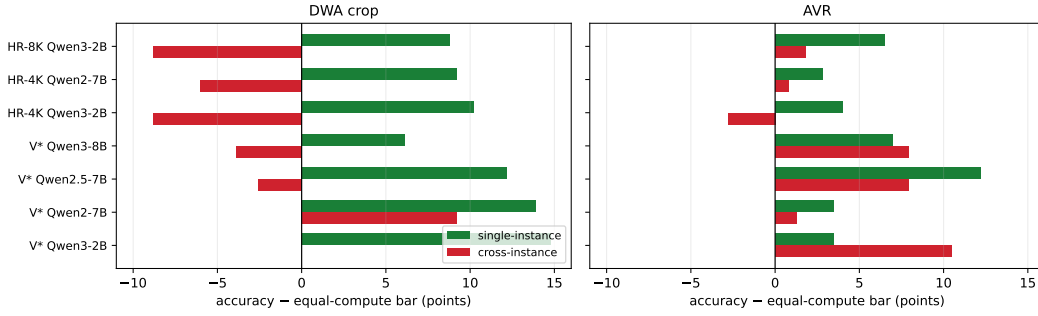


Figure 9: **The scope law.** Cropping is worth +6 to +15 points on single-instance questions in all seven cells and is negative on cross-instance in six of them. AVR is positive on both strata in six of seven, and gives up most of the single-instance gain to get there. The two policies are not variants of one method.

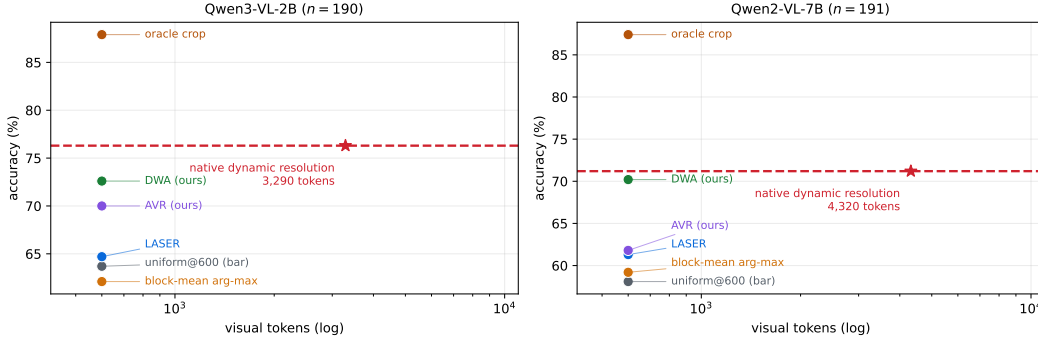


Figure 10: **Accuracy against visual-token compute, V*Bench.** The dashed line is the model run at its own native dynamic resolution. Our two policies sit on it at 14–18% of its compute; the published rules sit significantly below it; and a perfectly placed 600-token crop sits well above it. Placement is worth more than 5–7× the tokens.

H SCOPE AND COMPUTE, AS FIGURES

I MECHANISM: THREE STAGES, THREE INTERVENTIONS

Each row of Table 12 is measured by a different intervention, so no stage rests on the method that found another.

Table 12: The three stages. Every number is measured on both core checkpoints.

stage	layers	measured by	evidence
image → text transport	L0–L16	attention masking; hidden-state patching	suffix masks cost nothing from L16 (0.014, 0.029); the patch effect falls 17× and 96× between L12 and L16
question-conditioned read-out	L17–L21	end-task accuracy by read-out depth	crop accuracy spans 34.6–70.2, peaking at L17; no layer inside the window beats not cropping
answer tion	forma- L22+	logit lens	lens accuracy steps +18.0 [+8.7, +27.0] and +14.2 [+7.2, +21.3] above its early plateau

The logit lens is computed with the final norm applied to intermediate layers only; applying it to the last hidden state as well double-normalises and corrupts that layer alone, which cost us a retracted result earlier in this project. We assert per run that our last-layer read-out matches the true logits.

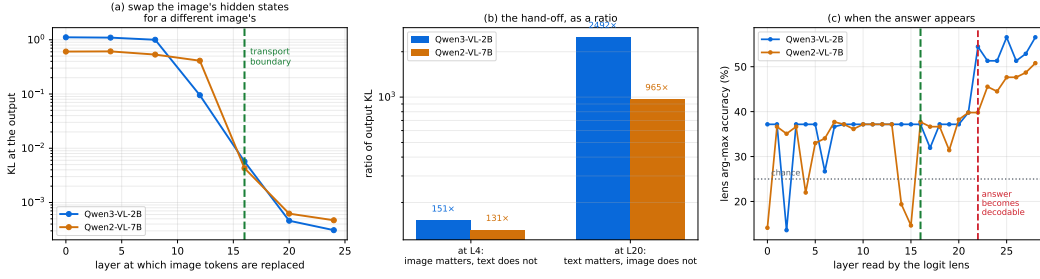


Figure 11: **(a)** Replacing the image tokens’ hidden states at layer ℓ with a different image’s: the output effect falls three orders of magnitude across the boundary. **(b)** The hand-off as a ratio: early, the image dominates and the text is inert; late, exactly the reverse. **(c)** Logit lens at the answer position: the correct option becomes decodable only in the last quarter of the stack.

A fourth, independent technique places the read-out band in the same spot: a ridge probe from the answer-position state to the target’s location rises across the transport window and peaks at L18–19 (Qwen3, $r = +0.635$) and L18 (Qwen2, $+0.603$).

J PROPOSITIONS, AND WHAT VALIDATES THEM

We state four theorems, three propositions and two lemmas. They are not deep results — two are tail-dominance and sufficiency statements, two are information bounds, and the lemmas are one-line linear algebra — but each makes a prediction that could have failed on data, and we report where they hold approximately rather than exactly. All measurements are out-of-fold on both core checkpoints; $n = 126$ (modal grid) for coverage, $n = 191$ for end-task and distributional tests.

Setup. For item i , layer ℓ , cell c , let $A_\ell^{(i)}(c) \geq 0$ be the last-prompt-token attention mass on cell c , row-normalised so $\sum_c A_\ell^{(i)}(c) = 1$. Write $m_\ell = \mathbb{E}_i A_\ell^{(i)}$ for the *item-mean* (item-independent) map and $A_\ell^{(i)} = m_\ell + U_\ell^{(i)}$, so $\mathbb{E}_i U_\ell^{(i)} = 0$.

Theorem 1 (Tail dominance of the conventional read-out). *Write each depth’s per-cell score as $x_\ell(c) = m_\ell(c) + u_\ell(c)$ with u_ℓ zero-mean over items, and put $\Delta_B = \max_c \sum_{\ell \in B} m_\ell(c) - \text{med}_c \sum_{\ell \in B} m_\ell(c)$. The block read-out returns $\arg \max_c \sum_{\ell \in B} x_\ell(c)$.*

- (i) *If Δ_B exceeds the range of $\sum_{\ell \in B} u_\ell$, the arg-max is $\arg \max_c \sum_B m_\ell$ for every item: a fixed cell independent of the image.*
- (ii) *Equal-weight positive averaging cannot escape (i) at any band width: the m_ℓ leads are sign-aligned across depth, so Δ_B is superadditive in B .*
- (iii) *Any per-cell rescaling that equalises m across cells — in particular dividing by an estimate of m — removes the dominance without using labels.*

Proof. (i) If the deterministic part’s lead exceeds the total variation the item-specific part can supply, it decides the maximum. (ii) Each summand contributes its m_ℓ lead with the same sign, so Δ_B is superadditive; whether Δ_B/σ_B rises or falls depends on the covariance of the u_ℓ , and the claim is only that it does not reach the $O(1)$ regime in which $\sum_B u_\ell$ could decide the arg-max. (iii) Immediate from (i) applied to the rescaled scores. $\square$

Lemma 1 (Leakage decomposition). *For every weight vector w , $\sum_\ell w_\ell A_\ell^{(i)} = L(w) + \sum_\ell w_\ell U_\ell^{(i)}$ with $L(w) = \sum_\ell w_\ell m_\ell$ independent of i . The item-independent component of the score is exactly $L(w)$.*

Proof. Substitute the decomposition and rearrange. $L(w)$ carries no i -dependence, so its ranking of cells is identical for every item. $\square$

Lemma 2 (Non-negative filters cannot cancel a common prior). *If the item-mean maps are rank-one across depth, $m_\ell = \alpha_\ell \mu$ with $\mu \neq 0$ and $\alpha_\ell > 0$, then $L(w) = (\alpha^\top w) \mu$. Hence $L(w) = 0$ iff $\alpha^\top w = 0$; no non-zero $w \geq 0$ achieves zero leakage.*

Proof. Linearity, then $\alpha^\top w = \sum_\ell \alpha_\ell w_\ell > 0$ whenever $w \geq 0$, $w \neq 0$. $\square$

Every part is measured, and the measurements decide between accounts that (ii) alone cannot separate. On the items the block mean gets wrong, the cell it selects sits at the **100th percentile** of the item-independent map — on both checkpoints, on every failure. It is not mis-weighting evidence; it is returning the raster sink. Consistent with (ii), Δ_B/σ_B measured over band widths 1 to 11 stays between 10.2 and 16.6 on one checkpoint and between 7.0 and 15.1 on the other — always an order of magnitude above the regime where the item-specific term could win, so no amount of averaging rescues the rule. The ratio *rises* with width on the first checkpoint and *falls* on the second, so we claim only that averaging never escapes, not that wider is monotonically worse. What is common to both is that the deployed eleven-wide band is worse than the best narrower band (0.418 $\rightarrow$ 0.083 and 0.390 $\rightarrow$ 0.297), and that its last-column rate is higher there (17.5 $\rightarrow$ 91.3% and 45.2 $\rightarrow$ 62.7%). Consistent with (iii), an *unsupervised* correction — divide each cell by its value on a typical item, then take the arg-max — lifts coverage from 0.083 to 0.408 and from 0.292 to 0.504, reaching 71% and 86% of what the fitted read-out achieves, and drops the last-column rate to 2.4% and 0.0% against a 4.8% chance rate, with no ring mask.

And it buys nothing at the end task, which bounds what this proposition explains. Run as a placement rule and answered, the same label-free correction scores 59.7% and 61.3% against the block mean’s 62.3% and 59.2%: -2.6 $[-7.9, +2.6]$ and $+2.1$ $[-3.1, +7.3]$, neither clear of zero, and it does not beat not cropping at all. DWA beats it by $+13.1$ $[+5.8, +20.4]$ and $+8.9$ $[+2.6, +15.2]$, clear on both checkpoints. So the proposition explains why the conventional read-out *places* badly, and a label-free fix repairs the placement, but repairing placement is not what the fitted read-out is selling. Its end-task advantage survives after the placement failure has been removed by other means, and we do not yet have a mechanism for that remainder. This is also the largest coverage-to-accuracy gap we have measured — 66–70% of the coverage gap closed, none of the accuracy gap — and it is a caution that applies to our own use of coverage in this section, not only to prior work.

What the fit adds, and why its sign does not matter. Because the features are log-attention, a weight vector summing to zero computes a *ratio* across depths, which cancels any per-cell factor common to all layers; the fitted weights are close to that, with $|\sum_\ell w_\ell|/\sum_\ell |w_\ell| = 0.126$ and 0.225. But that is not the only route. Sink dominance itself varies $3.2\times$ and $4.4\times$ across depth, and the least-dominated depths are exactly the read-out band, so a strictly non-negative read-out can escape by *selecting* those depths rather than by cancelling anything. Both routes are available, which is why constraining every weight to be non-negative costs nothing while zeroing 44 and 43 of 56 coefficients. The item-mean maps are approximately rank-one (leading eigenvalue share 0.86 and 0.77), and in map space the non-negative constraint costs only 0.044 and 0.002 mean coverage: the sign of the fitted weights is an artefact of ridge collinearity between adjacent layers, not the mechanism. We do not claim the fitted weights track dominance depth-by-depth: that correlation is only $+0.171$ and $+0.131$. The supported statement is that the failure is tail dominance, that it has an unsupervised fix which recovers most of the gap, and that supervision buys the remaining 14–29%.

A refinement the theory predicted, which failed. If what remains after dividing out m were heteroscedastic across cells, the right label-free statistic would be the studentised one, $(x(c) - m(c))/s(c)$. We pre-registered that it would beat the plain ratio. It does not, on either checkpoint (-0.015 and -0.028), and it closes 63% and 61% of the gap against the plain ratio’s 66% and 70%. The proposition therefore explains the failure and supplies a label-free correction, but does not yet predict a better one: the residual that supervision buys is not simple heteroscedasticity.

Two accounts we tested first and rejected. That signed weights cancel a common nuisance: refuted, since non-negative fits lose nothing and the orthogonality the argument needs measures $+0.431$ and $+0.388$ rather than zero. That the fit is a matched filter over depth, weighting by per-layer signal-to-noise: refuted, one of two on each of its predictions, and wrong in kind — the block mean already retains 77% of attainable signal-to-noise on both models while its coverage is several times worse, and a mean-based quantity cannot explain an arg-max failure.

Theorem 2 (A single-depth read-out is lossy unless that depth is sufficient). *Let c^* be the cell containing the evidence and $A = (A_1, \dots, A_L)$ the depth profile. For any rule $\hat{c} = g(A_\ell)$ reading a single layer: (i) $c^* \rightarrow A \rightarrow A_\ell \rightarrow \hat{c}$ is a Markov chain, so $I(c^*; \hat{c}) \leq I(c^*; A_\ell) \leq I(c^*; A)$; (ii) the right inequality is an equality iff $c^* \perp A \mid A_\ell$, i.e. A_ℓ is sufficient for c^* within the profile; (iii) by Fano, $\Pr[\hat{c} \neq c^*] \geq (H(c^*) - I(c^*; A_\ell) - 1)/\log |C|$.*

Proof. A_ℓ is a coordinate projection of A and $\hat{c}$ is post-processing of A_ℓ , so the chain holds and the data-processing inequality applies along each link, giving (i). For (ii), $I(c^*; A) - I(c^*; A_\ell) = I(c^*; A \mid A_\ell) \geq 0$ by the chain rule, with equality exactly under conditional independence. (iii) is Fano applied to that chain. $\square$

Theorem 2 is vacuous until sufficiency is decided, and sufficiency is an empirical property of a given model, so we test its equality condition. If A_ℓ were sufficient its arg-max would be correct wherever any layer’s is, so the best single depth would match the family’s envelope. Cropping at every one of the 28 layers’ arg-max and answering gives:

Table 13: The sufficiency test at the end task. A sufficient layer would make the first two columns equal.

	best depth	some depth	gap	every depth	spread over depth
Qwen3-VL-2B	70.2	95.3	+25.1	5.2	34.6–70.2
Qwen2-VL-7B	62.8	89.5	+26.7	9.9	36.1–62.8

Sufficiency is rejected on both checkpoints by 25 and 27 points. The escape is to read every depth: DWA is +2.6 [−3.7, +8.9] and +7.3 [+1.0, +13.1] against the best single depth — a depth chosen *on the test set*, so an optimistic baseline — clear on one checkpoint and matching on the other, while the published fixed choice sits 31.9 and 19.9 points below that oracle. The scope of the test is stated: the envelope is measured over the arg-max read-out family, which is the class a depth-selection rule chooses among, not over all measurable functions of A .

Theorem 3 (Decoding bound, and a certified gap for the conventional read-out). *For any read-out decision $\hat{c}$ of a per-cell score and target cell T : (i) $I(\hat{c}; T) \leq H(\hat{c}) \leq \log |C|$; (ii) $I(\hat{c}; T) \geq H(T) - H_b(P_e) - P_e \log(|C| - 1)$ with $P_e = \Pr[\hat{c} \neq T]$; (iii) if $\hat{c}$ is constant then $I(\hat{c}; T) = 0$.*

Proof. (i) $I(\hat{c}; T) = H(\hat{c}) - H(\hat{c} \mid T) \leq H(\hat{c}) \leq \log |C|$. (ii) Fano’s inequality. (iii) $H(\hat{c}) = 0$, so (i) gives $I = 0$. $\square$

The bounds are exact and directly measurable, and the same maps admit an oracle decision that *is* the target, so $I(\text{oracle}; T) = H(T) = 6.06$ bits on the modal-grid items. The conventional read-out does not reach that: the literal block mean (no ring mask) has decision entropy 1.02 and 3.46 bits on the two checkpoints and lands in the grid’s last column on 84.8% and 61.8% of the full item set, so it can carry at most that much — a certified gap of ≥ 5.0 and ≥ 2.6 bits. The ring mask restores the baseline’s decision entropy (6.30/4.60 bits) but not its accuracy; the fitted read-out attains the same entropy with higher coverage (0.605/0.581 against 0.459/0.402). The failure is in the decoder, not the representation. Two consequences. The information a decision can carry is raised only by making it *varied* or *accurate* — the bounds never mention cancellation — so a non-negative depth fit reaches the same terms by selecting the depths where the item-independent component is least dominant (§6.1). And the depth *ordering* changes both terms while the sign of the weights does not.

Theorem 4 (The information horizon). *Suppose the visual-token values after layer p are output-invariant: $I(V_{>p}; Y \mid \text{state}_p) = 0$ (value-inertness). Then (i) pruning to any keep-set loses no answer information; (ii) AVR’s gain over the bar E_0 is the resolution information gain $\text{acc}(E) - \text{acc}(E_0)$, bounded by $I(X; Y) - I(Z_{E_0}; Y)$; and (iii) the information-bearing depth is $p + 1$ layers, so the affordable resolution multiplier is $N/(p + 1) = 1.647\times$ over uniform-depth encoding.*

Proof. (i) The premise makes the answer conditionally independent of those values given the state at p , and the keep-set is a function of them. (ii) Pruning is information-free by (i), so AVR and uniform encoding at E induce the same distribution; the data-processing inequality bounds the remaining

difference by the information the bar’s encoding fails to carry. (iii) Token-layer accounting with the information-bearing depth. $\square$

The premise is measured with its violation bounded by Pinsker, $TV \leq \sqrt{KL/2}$: at the boundary the answer distribution depends on the token values by at most 0.033–0.044 in total variation, while the same patch one window earlier moves it by 0.36–0.41 (Appendix I). The horizon is sharp, and it is the same $0.57\times$ depth the masking experiments locate. The conclusion is what Appendix F tests: gain tracks headroom with $r = 0.966$ over ten cells, and again with $r = 0.911$ over eight further cells whose headroom is near zero, where it correctly predicts that AVR does nothing. *Where it is only approximate*: the premise holds to $KL \approx 5 \times 10^{-4}$, not to zero, so the conclusion is approximate too. Measured slippage is at most 0.5 points in seven of ten cells and -1.8 in one. The theorem’s real content is that the *choice* of keep-set is irrelevant, which is why random keep matches attention keep on 96.3% of items.

Proposition 1 (The budget determines the affordable resolution). Pruning to a fraction k at layer P of N costs $TL = E(P+1) + kE(N-1-P)$ token-layers. Matching a uniform bar of B tokens gives $E^* = BN/((P+1) + k(N-1-P))$. With $B = 600$, $N = 28$, $P = 16$, $k = 0.1$ this is $E^* = 928$, a $1.55\times$ resolution multiplier. We ran $E = 900$, 97% of the optimum, and realised 96–100% of the bar. This is arithmetic; we state it because it is the design equation, and because it says the multiplier is fixed by the boundary’s depth fraction rather than chosen.

Proposition 2 (Value-inertness implies selection irrelevance). Suppose that for every subset D of visual positions, replacing the hidden values of D at layer p with arbitrary donor values leaves the option distribution unchanged. Then for any two keep-sets D_1, D_2 of equal size, the models $U_E \circ P_{D_1}$ and $U_E \circ P_{D_2}$ induce the same distribution: every ranking rule is equivalent.

Proof. The premise makes the output invariant to the values of any subset of visual positions at layer p . The kept set is such a subset, so no choice of kept set can change the distribution. $\square$

Validation. On the unpruned 900-token pass at the boundary (L16, Qwen3, $n = 191$), replacing the hidden states of the 810 pruned positions with another image’s gives KL 0.0022 (1.0% flips) and replacing the 90 kept positions gives KL 0.0065 (3.1%); the same patch at L8 gives 0.341 (20.9%). Qwen2 replicates (0.0038/3.7% and 0.0027/0.5% at L16; 0.2622/29.8% at L8), so the premise holds on both models.

Proposition 3 (A single window cannot serve two distant sites). A window of side W contains at most one of two evidence sites separated by more than W . *Trivial, and it is not the interesting part. The interesting part is empirical: with perfect placement, cross-instance error stays at 25% and 22% against 3.5% and 6.1% for single-instance, and two full-resolution crops covering both objects still score -2.6 and -3.9 against one. The residual is not a perceptual failure, so no reallocation of perceptual resource reaches it — which is why the scope law is a property of the question type and not of the method.*

K ADDITIONAL VISUALISATIONS

L DESIGNS THAT DID NOT WORK

The last row generalises the rest, and explains why the two policies cannot be merged into one. DWA’s signal is produced after the transport boundary, and pruning only saves compute at or before it. A pipeline that prunes early enough to save has destroyed the layers DWA reads, which costs 3.1 and 12.6 points of coverage; a pipeline that waits for DWA has nothing left worth pruning. Holding the ranking fixed and moving only the cut from L16 to L4 costs 5.2 points. The incompatibility is structural, not a tuning failure.

The two policies are substitutes, not complements. The structural argument above is not the only reason they do not combine. We also tested the composition that *is* mechanically sound: hold the token-layer budget fixed, and let any placement rule spend its answer pass on a crop encoded at 460 tokens and pruned 90% at the boundary instead of a plain crop at 300. Applied to ViCrop-style block-mean, to LASER, to DWA and to oracle placement, on both checkpoints, every one of

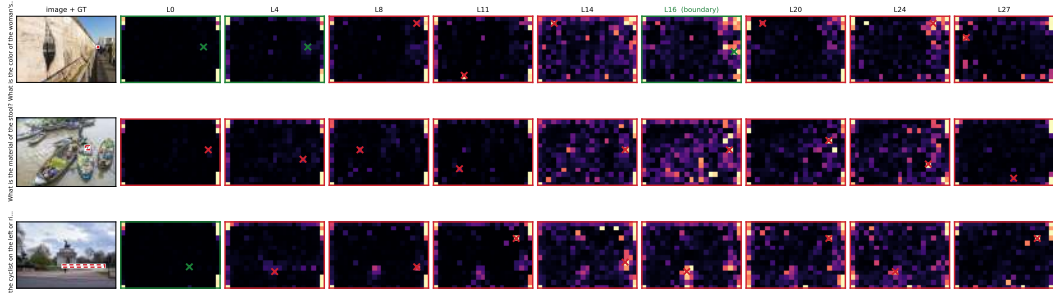


Figure 12: **The same image, the same question, a different answer about where to look at every depth.** Each panel is one layer’s attention over image cells (colour clipped at the 98th percentile so the serialisation sink does not wash out the structure); the cross is that layer’s ring-masked arg-max, green where the resulting crop would cover the evidence and red where it would not. Nothing about the input changes across a row. Only the read-out depth does.

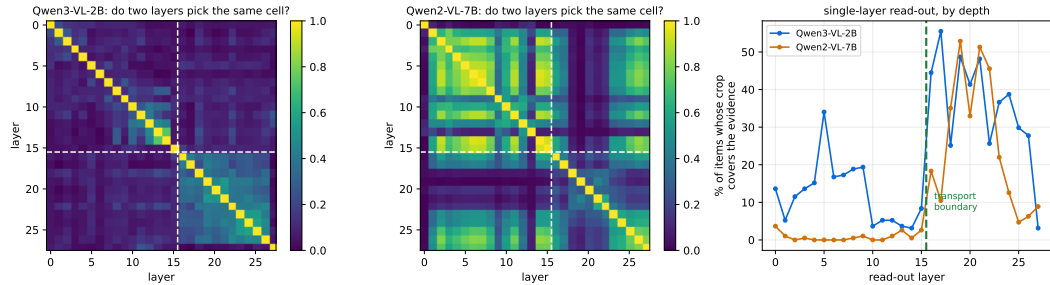


Figure 13: **Left, centre: two layers rarely choose the same cell.** Mean off-diagonal agreement is 13.8% and 38.1%; the block structure splits at the transport boundary (dashed). **Right: and only the layers after the boundary are worth reading.** The fraction of items whose single-layer crop covers the evidence is at or below 20% everywhere before the boundary, jumps immediately after it, and decays again toward the output. This is the paper’s central claim in one curve: the usable placement signal exists only after transport has finished.

the eight contrasts spans zero. The diagnostic says why: the resolution headroom left *on a crop* averages +0.8 points, against +4.5 on the full image. A 0.25×0.25 window re-encoded at 300 tokens has already lifted the object above the encoding cliff, so the magnification AVR would buy has already been spent. Both policies spend the same resource, effective resolution on the region that matters; cropping spends it in a targeted way and AVR spends it uniformly. That is why their *scope* is complementary while their *gains* are not additive.

Why cropping cannot reach cross-instance questions. Under *perfect* placement, cross-instance accuracy reaches 75.0 and 77.6, an irreducible error of 25% and 22%, against 3.5% and 6.1% for single-instance. Two crops at full resolution covering *both* objects, which raises cross-instance object coverage by 17.1 and 15.8 points, still score -2.6 and -3.9 against a single crop. Cross-instance error is not predominantly perceptual, so no reallocation of perceptual resources reaches it. Resolution does reach part of it, which is what the second policy exploits, but cropping does not.

M PIPELINE FAULTS FOUND, AND HOW THEY WERE CAUGHT

Two runs in this work produced plausible numbers that were wrong. We report them because the rules that caught them are the reason we trust the rest, and because both faults are easy to make and invisible in aggregate accuracy.

A silently unpatched module. Pruning is injected by monkey-patching the attention forward pass. The patch covered two module paths; a third checkpoint’s attention lived in a third module, so



Figure 14: **Five placement rules, same items, same budget.** Selection rule: single-instance items on which the oracle crop is correct, so the ceiling behaves like a ceiling, and on which DWA is correct while the fixed-layer rule is not; 61 items qualify and the first four by question id are shown. The fixed mid-layer recipe crops glass, masonry and sky.

pruning was a no-op and the arm was byte-identical to its unpruned control while reporting a clean +12.2. It was caught by a standing rule: a contrast of exactly +0.0 with a $[+0.0, +0.0]$ interval is a pipeline fault, never a result, because real effects do not land on zero to machine precision. The fix asserts at load time that the model’s actual attention module is one the patch reached, and prints it every run.

A hard-coded depth on a differently-shaped model. The arm table hard-coded the prune layer at 16. On the 36-layer checkpoint the correct depth, $0.57 \times 36 = 21$, was computed and printed but never used, so the run pruned at 44% of depth, inside the transport window. It was caught by the compute budget reading 78% of the bar instead of the expected 98%. We retain the voided run as accidental evidence for the boundary itself: pruning inside the window cost -7.9 pooled and -10.5 cross-instance on a third model family.

Every arm therefore prints its realised token count and its token-layer budget as a fraction of the bar, and any run outside 96–104% is discarded rather than reported.

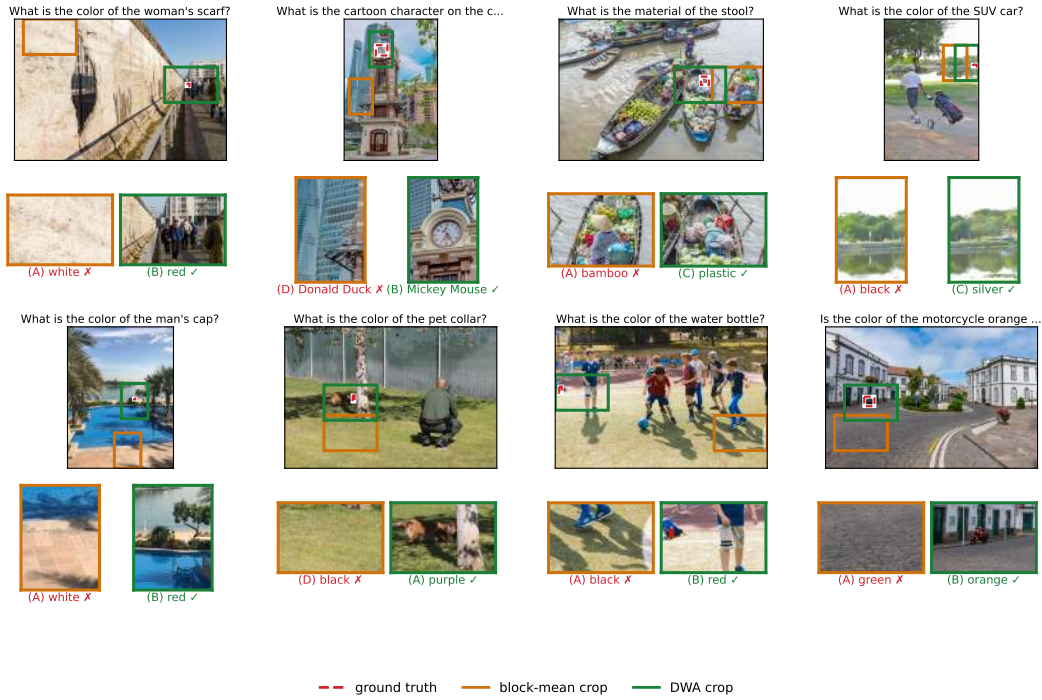


Figure 15: **Eight inferences, Qwen3-VL-2B.** Top of each cell: the image with the ground-truth box, the block-mean crop and the DWA crop. Below: the two crops the model actually answers from, with its answer.

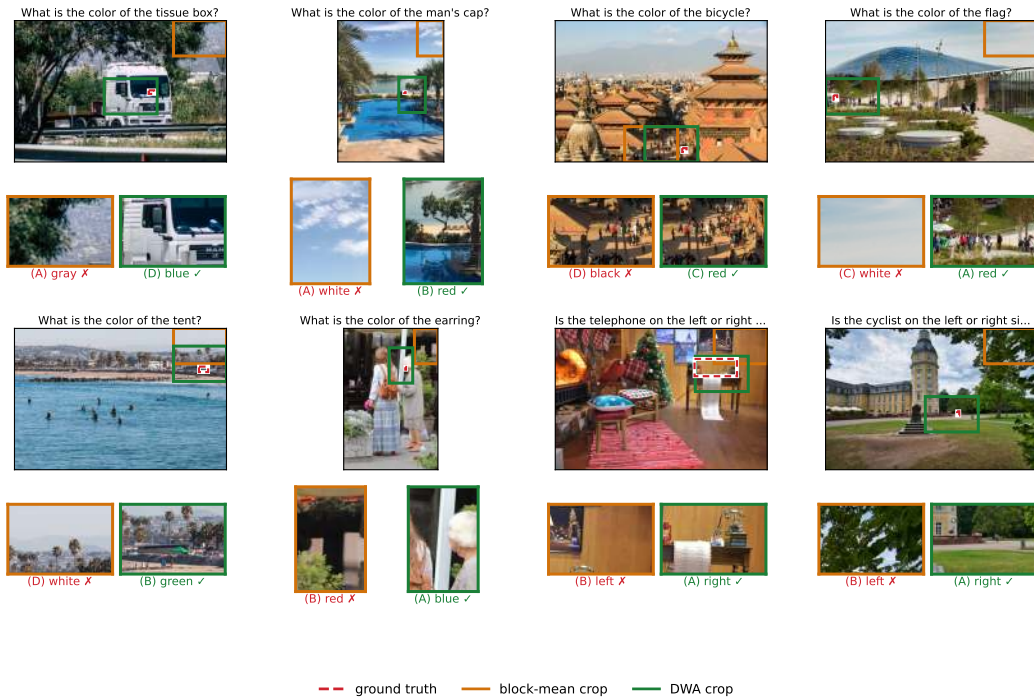


Figure 16: **The same eight-item view on Qwen2-VL-7B.** The failure mode is the same on a different model family and a different parameter count.

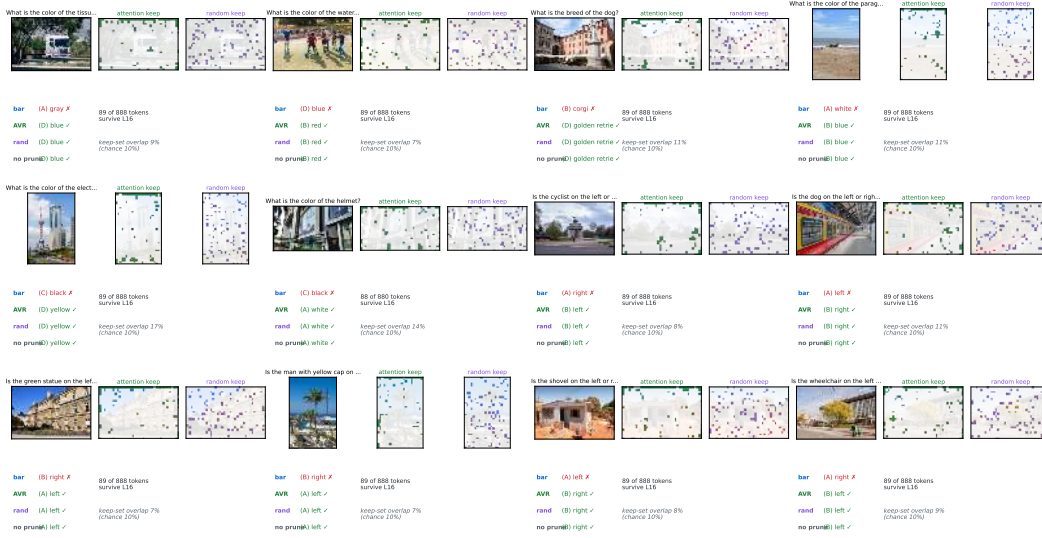


Figure 17: **Twelve inferences under AVR.** For each item: the image, the tokens kept by attention at the boundary, and an equally sized random keep. The two keep-sets overlap at close to the 10% chance rate and the model answers identically under both, and identically again without pruning at all.



Figure 18: **The serialisation sink, averaged over the benchmark.** Each item’s read-out map is resampled to a common grid and averaged over all 191 items. Mass concentrates in the last column — the token before a raster row-wrap — on both checkpoints, at roughly four times the uniform share. This is what the outer-ring mask in every deployed implementation is patching around.

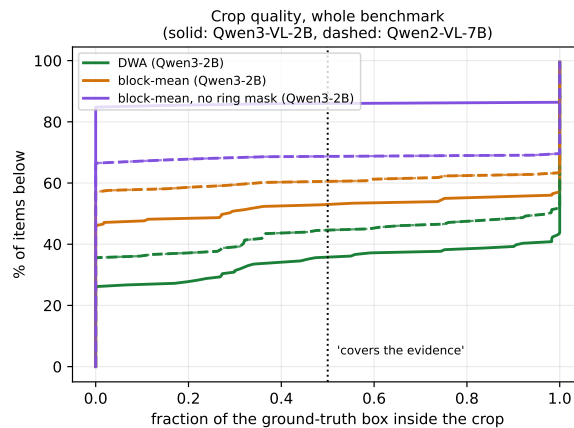


Figure 19: **Crop quality over the whole benchmark.** Cumulative distribution of the fraction of the ground-truth box captured by the crop. Without the ring mask the block-mean rule is close to useless; with it, it is usable; DWA is better than both and barely uses the mask.

Table 14: The ten designs intended to extend cropping to cross-instance questions, and the two that tried to combine the two policies. Each was pre-registered with a primary contrast and a guard before it ran. “Cleared” counts checkpoints of two on which the primary passed.

design	what it changed	cleared	diagnosed cause
per-item window sizer	crop size from map features	0/2	grid inversion, weak features
peak-spanning window	box spanning the top peaks	0/2	relative signal, absolute constant
mass-containment window	window holding $x\%$ of mass	0/2	boxes far too large
joint cell-window ranking	arg-max over both at once	0/2	metric artefact: coverage rises with size
expected-accuracy window	window by predicted accuracy	0/2	accuracy–size curve flat against noise
thumbnail in answer pass	crop plus 64-token scene	0/2	second object below the encoding cliff
global–local split	scene@150 plus crop@150	0/2	halving the crop costs more than the scene gives
context-continued crop	scene@300 plus crop@300	0/2	scene costs 8.7 and 5.2 even with a perfect crop
multi-crop, matched budget	2×150 and 3×100	0/2	magnification tax (-11.3 , -4.3) exceeds coverage gain ($+7.8$, $+2.6$)
funded multi-crop	2×300 , pruned, early exit	0/2	$+3.7$ on one model, -2.6 on the other
DWA-guided early pruning	rank at L4 instead of L16	0/2	-5.2 : ranking before transport is worthless
AVR stacked on a placement rule	crop@460 pruned at L16, same budget	0/2	no resolution headroom left on a crop ($+0.8$ vs $+4.5$)